

NEERAJ K

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EDUCATION

Degree	Institution	Score	Year
B.E. — Electronics & Communication Engineering	Easwari Engineering College, Chennai	8.2 / 10 (Current)	2024–Present
Senior Secondary (Class XII)	Chinmaya Vidyalaya Sr. Sec. School, Virugambakkam	90.8%	2024
Secondary (Class X)	Chinmaya Vidyalaya Sr. Sec. School, Virugambakkam	90.2%	2022

PROJECTS

4-Bit R-2R Ladder Digital-to-Analog Converter | Self-Guided Project

- Designed and implemented a 4-bit DAC using an R-2R resistor ladder network to convert binary inputs into proportional analog voltage levels.
- Validated output using Arduino and LM358 op-amp buffering; demonstrated accurate staircase waveform generation and confirmed DAC linearity across all 16 binary combinations.

Encoder-Based PID Motor Control System | Self-Guided Project

- Developed a closed-loop PID motor control system using Arduino Uno, encoder feedback, and BTS7960 motor driver for real-time speed and position regulation.
- Tuned PID parameters (Kp, Ki, Kd) to minimize overshoot and settling time; validated stable response under varying load conditions.

Automated Document Formatting and Structuring System | Self-Guided Project

- Built a browser-based tool to extract and restructure unformatted DOCX content into standardized formats using JavaScript and the docx library.
- Implemented real-time preview and instant file regeneration, reducing manual formatting effort significantly.

TECHNICAL SKILLS

- Programming Languages:** C, C++, Python
- Embedded Platforms:** Arduino Uno, Embedded C, Real-Time Control
- Hardware & Circuit Design:** R-2R DAC, PID Control, Sensor Fusion, Breadboard / PCB Prototyping, LM358 Op-Amp
- EDA Tools:** Tanner EDA, Altium 365
- Software / Tools:** Visual Studio Code, Jupyter Notebook, Google Colab
- Databases:** SQL
- Soft Skills:** Communication, Leadership, Teamwork, Creativity, Accountability

CERTIFICATIONS & KEY COURSES

- AI / ML:** Fundamentals of AI — Cisco Networking Academy & Qualcomm Academy
- Computer Architecture:** RISC-V Introduction — RISC-V International
- Programming:** C and C++ Programming — Orange Institute

EXTRACURRICULAR & ACTIVITIES

- Symposium Participation:** Designed a fire-extinguishing robot in AutoCAD; presented mechanical layout and functional deployment concepts at a college-level symposium.
- Hands-on Learning:** Self-driven projects in embedded systems, IoT integration, and sensor interfacing outside of coursework.